

PRODUCTION-READY AI AGENTS ON AZURE · LAB ENVIRONMENT

Building the lab environment by hand in the Azure portal

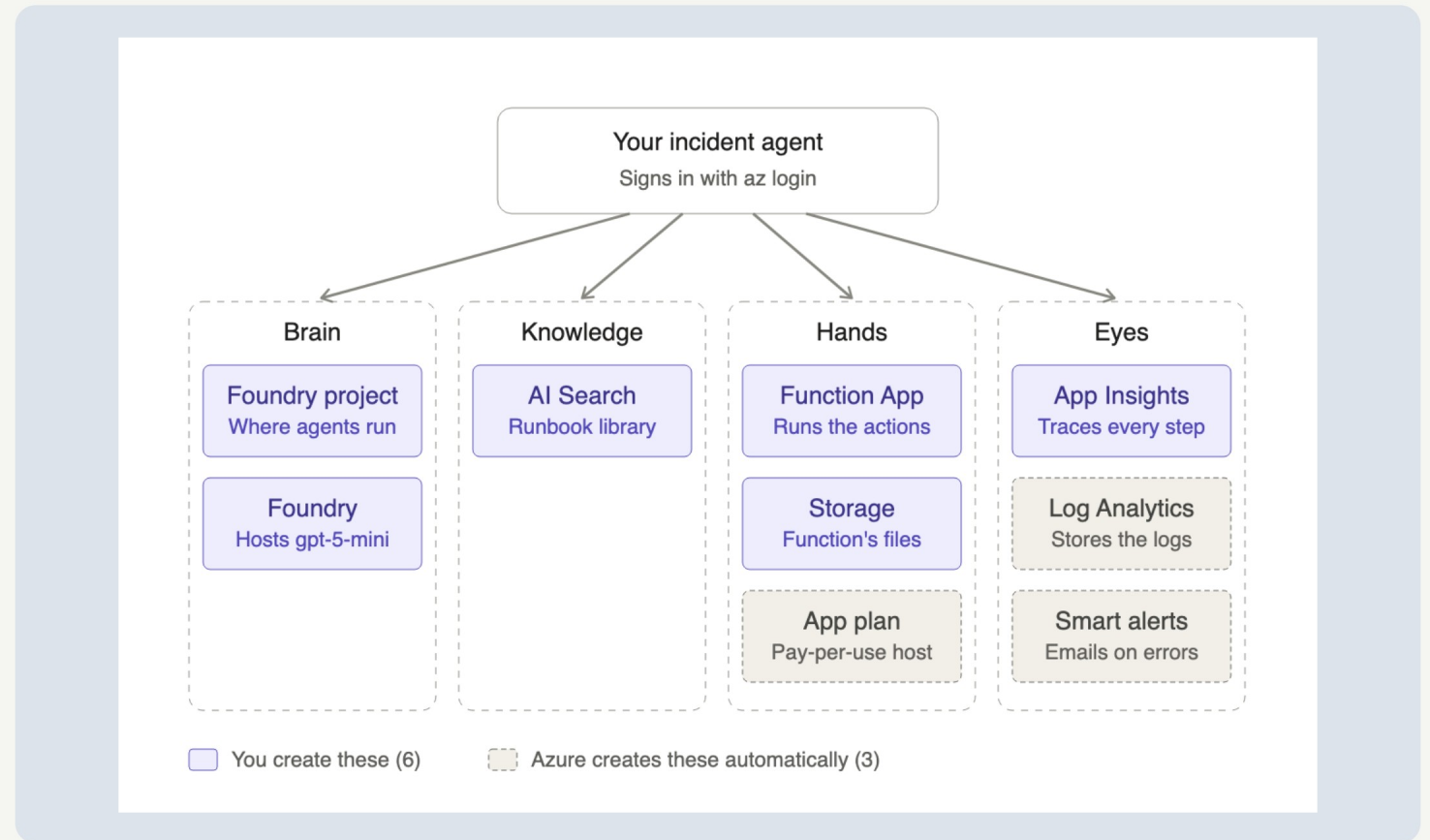
Every click, with the real screenshots: Foundry and models, AI Search, Storage, Application Insights, Function App, permissions and the .env file.

OVERVIEW

What you will build

Think of the agent as a new support-desk employee. It needs a **brain**, a **library** of approved runbooks, **hands** to take actions and **eyes** that record what it did.

You create six resources yourself.
Azure adds three more automatically (grey, dashed).



OVERVIEW

The order that works

#	What	Name we used	Region
1	Resource group (the folder)	rg-ai-agent-demo	East US 2
2	Foundry resource + project	aif-agent-demo-16555 / proj-default	East US 2
3	Two model deployments	gpt-5-mini · text-embedding-3-small	—
4	Foundry permissions	Foundry User · Cognitive Services User	—
5	AI Search + its settings	srch-agent-demo-16555 (Basic)	East US
6	Search permissions	Search Service + Index Data Contributor	—
7	Storage account	stagentdemo16555	East US 2
8	Application Insights	appi-agent-demo-16555	East US 2
9	Function App (Flex Consumption)	func-agent-demo-16555	East US 2
10	Keys, identity and the .env file	on the client machine	—

Four rules that prevent most mistakes

1

Check the subscription ID

Every create screen: it must start with 01ebc39d. A second account can have a subscription with the same name.

2

One folder, one region

Everything in rg-ai-agent-demo, East US 2. Only Search went to East US.

3

Two different search boxes

The big blue top bar finds services. The small box on the left finds settings inside the resource you are in.

4

Click Create or Deploy once

A second click makes a duplicate. On review pages, “(New)” next to something you already made means go back and pick yours.

STEP 2 · FOUNDRY

Create the Foundry resource and project

1. Portal top bar: search **Foundry**, open it, click **+ Create**.
2. Fill in the settings on the right.
3. Leave Storage, Key Vault and Application Insights empty.
4. **Review + create**, then **Create**. Both the resource and its project are created.
5. Open the project, **Overview**, copy the **Project endpoint**.

Tip

Name the project agent-demo-project if you want it to match the lab scripts exactly. proj-default also works; only the endpoint changes.

Setting	Value
Resource group	rg-ai-agent-demo
Name	aif-agent-demo-16555
Region	East US 2
Default project name	proj-default
Inbound networking	All networks
Outbound networking	No outbound networking

Resulting FOUNDRY_ENDPOINT

```
https://aif-agent-demo-16555.services.ai.azure.com  
/api/projects/proj-default
```

STEP 3 · MODELS

Deploy gpt-5-mini with custom settings

- 1. Project, Go to Foundry portal, then Build → Models → Deployments.
- 2. Deploy ▾ → Deploy a base model. Not “fine-tuned”: that list is empty.
- 3. Search **gpt-5-mini**. Not gpt-5.4-mini, gpt-5.6 or gpt-6.
- 4. Deploy ▾ → Custom settings, fill in the table, click Deploy once.

Field	Value
Deployment name	gpt-5-mini
Deployment type	Global Standard
Model version	2025-08-07
Tokens per minute	100000
Guardrails	DefaultV2

Watch out

The default was 1,000 tokens per minute: far too low for a class. Raise it with Edit. Clicking Deploy twice created gpt-5-mini-2; delete the duplicate, because both share one quota.

STEP 3 · MODELS

Deploy the embedding model

- 1. **Deploy** ▾ → **Deploy a base model** again.
- 2. Search **text-embedding-3-small**. Not -large, not ada-002: the search indexes are built for -small.
- 3. **Custom settings**, fill in the table, **Deploy** once.
- 4. The Deployments list should show exactly two rows, both Succeeded.

Field	Value
Deployment name	text-embedding-3-small
Deployment type	Global Standard
Model version	1
Tokens per minute	500000 (own quota)

Deployment	Status
gpt-5-mini · 100K TPM	Succeeded
text-embedding-3-small · 500K TPM	Succeeded

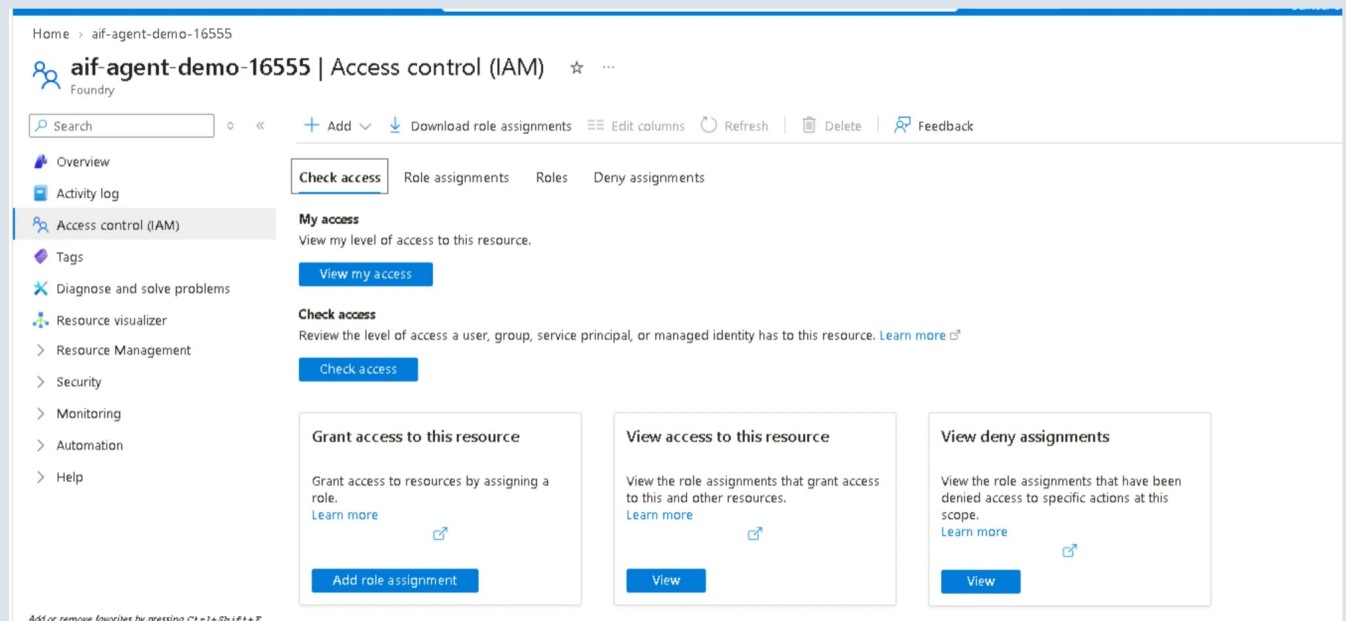
STEP 4 • FOUNDRY PERMISSIONS

Open Access control on the Foundry resource

1. Top bar: open **aif-agent-demo-16555**.
Type must say **Foundry**, not the project.
2. Left menu: **Access control (IAM)**.
3. Click **Add role assignment**.
4. You will add two roles for your own account.

Tip

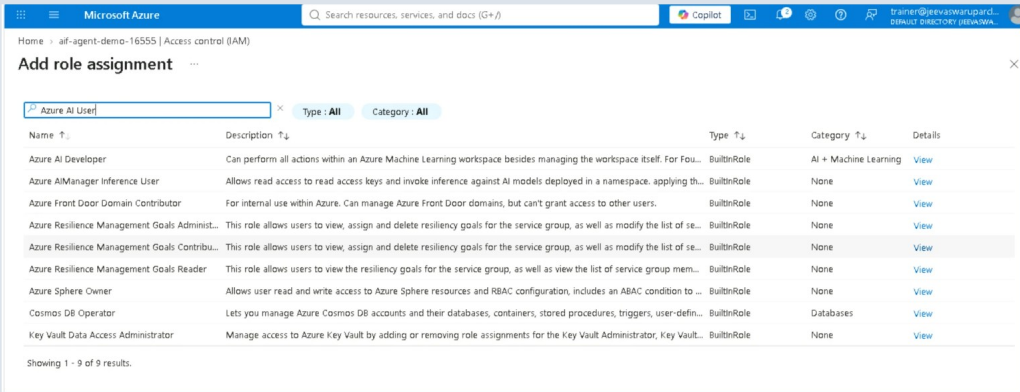
Owning a resource is not the same as permission to call its models or create agents. That needs data roles.



STEP 4 · FOUNDRY PERMISSIONS

“Azure AI User” is not offered here: use Foundry User

Search “Azure AI User”: not in the filtered list

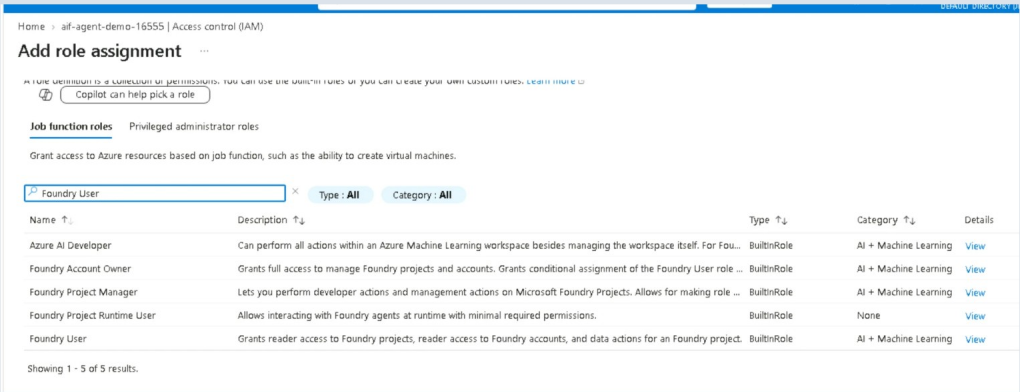


The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The search bar at the top contains the text 'Azure AI User'. Below the search bar, there are filters for 'Type: All' and 'Category: All'. The table below lists 9 roles, none of which are 'Azure AI User'.

Name	Description	Type	Category	Details
Azure AI Developer	Can perform all actions within an Azure Machine Learning workspace besides managing the workspace itself. For Fou...	BuiltInRole	AI + Machine Learning	View
Azure AIManager Inference User	Allows read access to read access keys and invoke inference against AI models deployed in a namespace, applying th...	BuiltInRole	None	View
Azure Front Door Domain Contributor	For internal use within Azure. Can manage Azure Front Door domains, but can't grant access to other users.	BuiltInRole	None	View
Azure Resiliency Management Goals Administ...	This role allows users to view, assign and delete resiliency goals for the service group, as well as modify the list of se...	BuiltInRole	None	View
Azure Resiliency Management Goals Contribu...	This role allows users to view, assign and delete resiliency goals for the service group, as well as modify the list of se...	BuiltInRole	None	View
Azure Resiliency Management Goals Reader	This role allows users to view the resiliency goals for the service group, as well as view the list of service group mem...	BuiltInRole	None	View
Azure Sphere Owner	Allows user read and write access to Azure Sphere resources and RBAC configuration, includes an ABAC condition to ...	BuiltInRole	None	View
Cosmos DB Operator	Lets you manage Azure Cosmos DB accounts and their databases, containers, stored procedures, triggers, user-defin...	BuiltInRole	Databases	View
Key Vault Data Access Administrator	Manage access to Azure Key Vault by adding or removing role assignments for the Key Vault Administrator, Key Vault...	BuiltInRole	None	View

Showing 1 - 9 of 9 results.

Search “Foundry User”: pick the last row



The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The search bar at the top contains the text 'Foundry User'. Below the search bar, there are filters for 'Type: All' and 'Category: All'. The table below lists 5 roles, with 'Foundry User' being the last row.

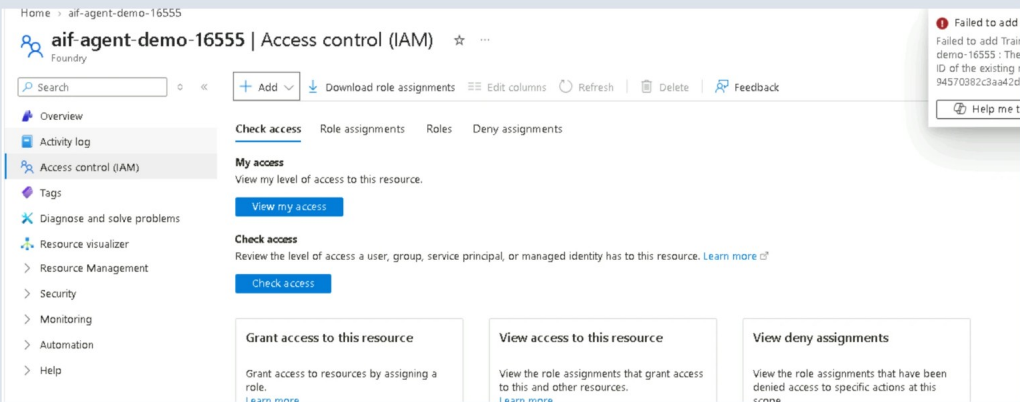
Name	Description	Type	Category	Details
Azure AI Developer	Can perform all actions within an Azure Machine Learning workspace besides managing the workspace itself. For Fou...	BuiltInRole	AI + Machine Learning	View
Foundry Account Owner	Grants full access to manage Foundry projects and accounts. Grants conditional assignment of the Foundry User role ...	BuiltInRole	AI + Machine Learning	View
Foundry Project Manager	Lets you perform developer actions and management actions on Microsoft Foundry Projects. Allows for making role ...	BuiltInRole	AI + Machine Learning	View
Foundry Project Runtime User	Allows interacting with Foundry agents at runtime with minimal required permissions.	BuiltInRole	None	View
Foundry User	Grants reader access to Foundry projects, reader access to Foundry accounts, and data actions for an Foundry project.	BuiltInRole	AI + Machine Learning	View

Showing 1 - 5 of 5 results.

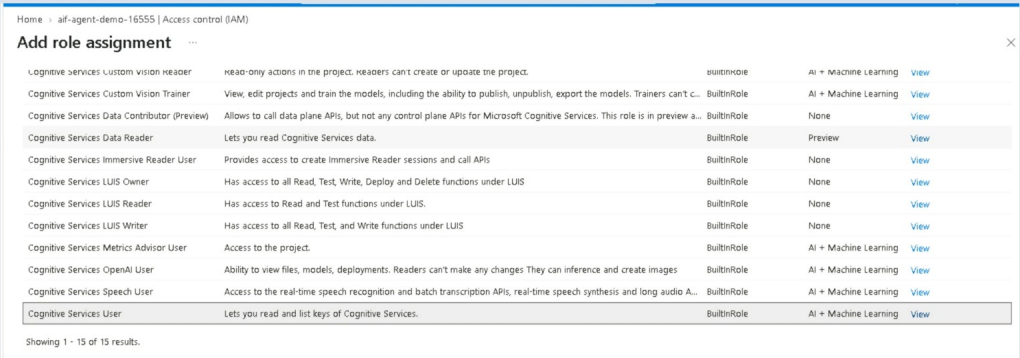
STEP 4 • FOUNDRY PERMISSIONS

“Already exists” is good news; then add Cognitive Services User

Foundry User already existed: nothing to do



Pick exactly “Cognitive Services User”



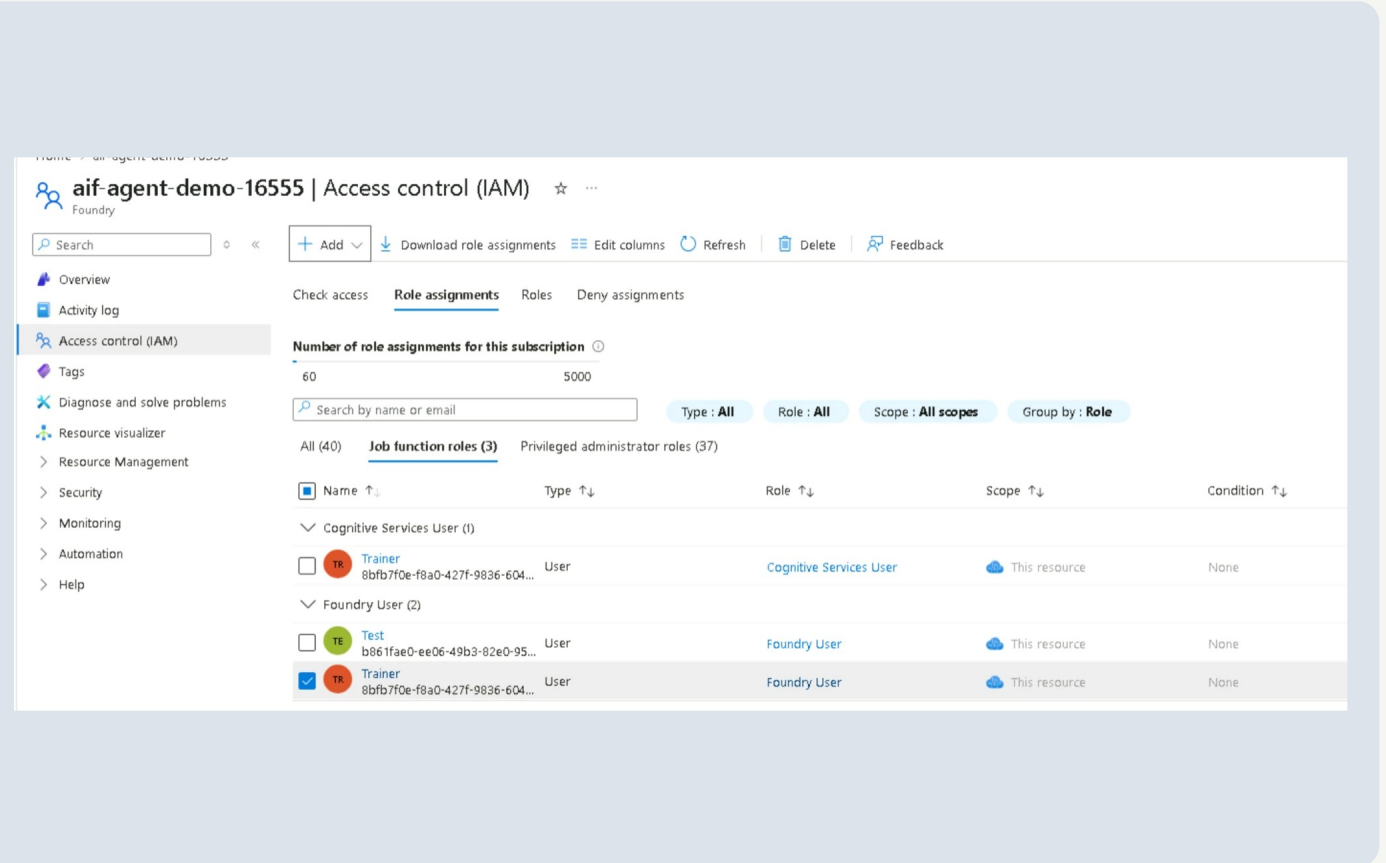
STEP 4 • FOUNDRY PERMISSIONS

Check: three job function roles

1. **Role assignments** tab, then **Refresh**.
2. Open **Job function roles**: it shows **(3)**.
3. Trainer has **Cognitive Services User** and **Foundry User**.
4. Untick any checkbox before you leave the page.

Watch out

While a row is ticked, the Delete button is live.
One wrong click removes your access.



STEP 5 · AI SEARCH

Create the Search service (the runbook library)

- 1. Top bar: **AI Search**, then **+ Create**.
- 2. Pricing tier: **Change pricing tier** → **Basic**. Not Standard (10× the price), not Free.
- 3. Scale 1 replica, 1 partition. Networking **Public**.
- 4. **Review + create**, then **Create**.

Watch out

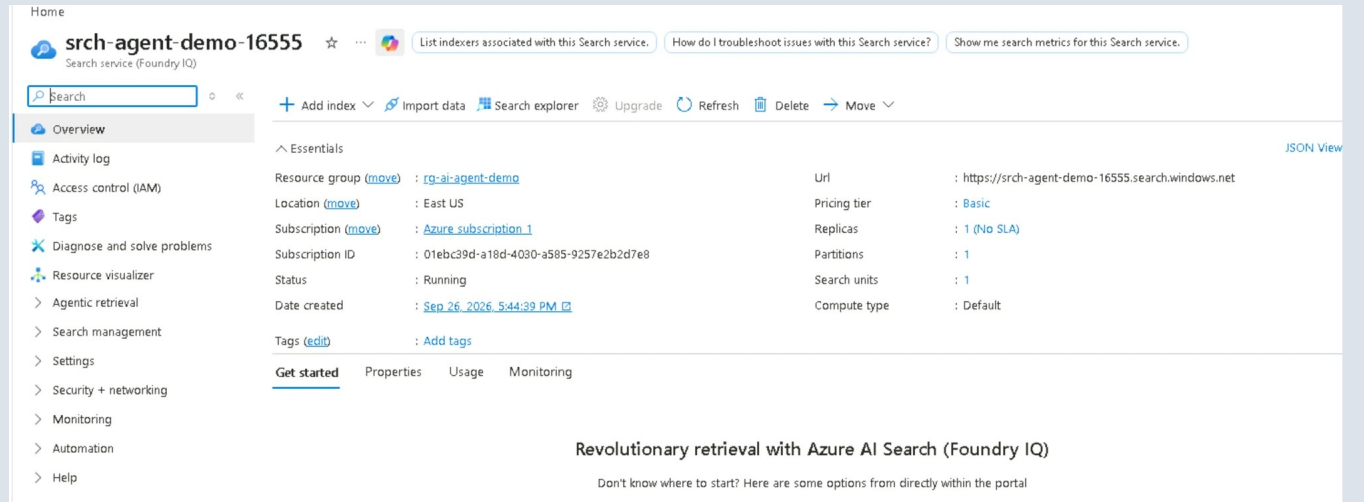
East US 2 answered “out of the resources required to provision new services”. Nothing is created and nothing is charged; create it again in **East US**. Other fallbacks: Central US, West US 2.

Setting	Value
Resource group	rg-ai-agent-demo
Service name	srch-agent-demo-16555
Location	East US
Pricing tier	Basic
Estimated cost	\$75/month ≈ \$2.50/day

STEP 5 · AI SEARCH

Copy the Search address

1. Go to resource.
2. Check: **Running, Basic, East US**, subscription **01ebc39d....**
3. Copy the **Url**: it becomes **SEARCH_ENDPOINT** in **.env**.
4. The service starts **empty**. Indexes are built later from the client machine.



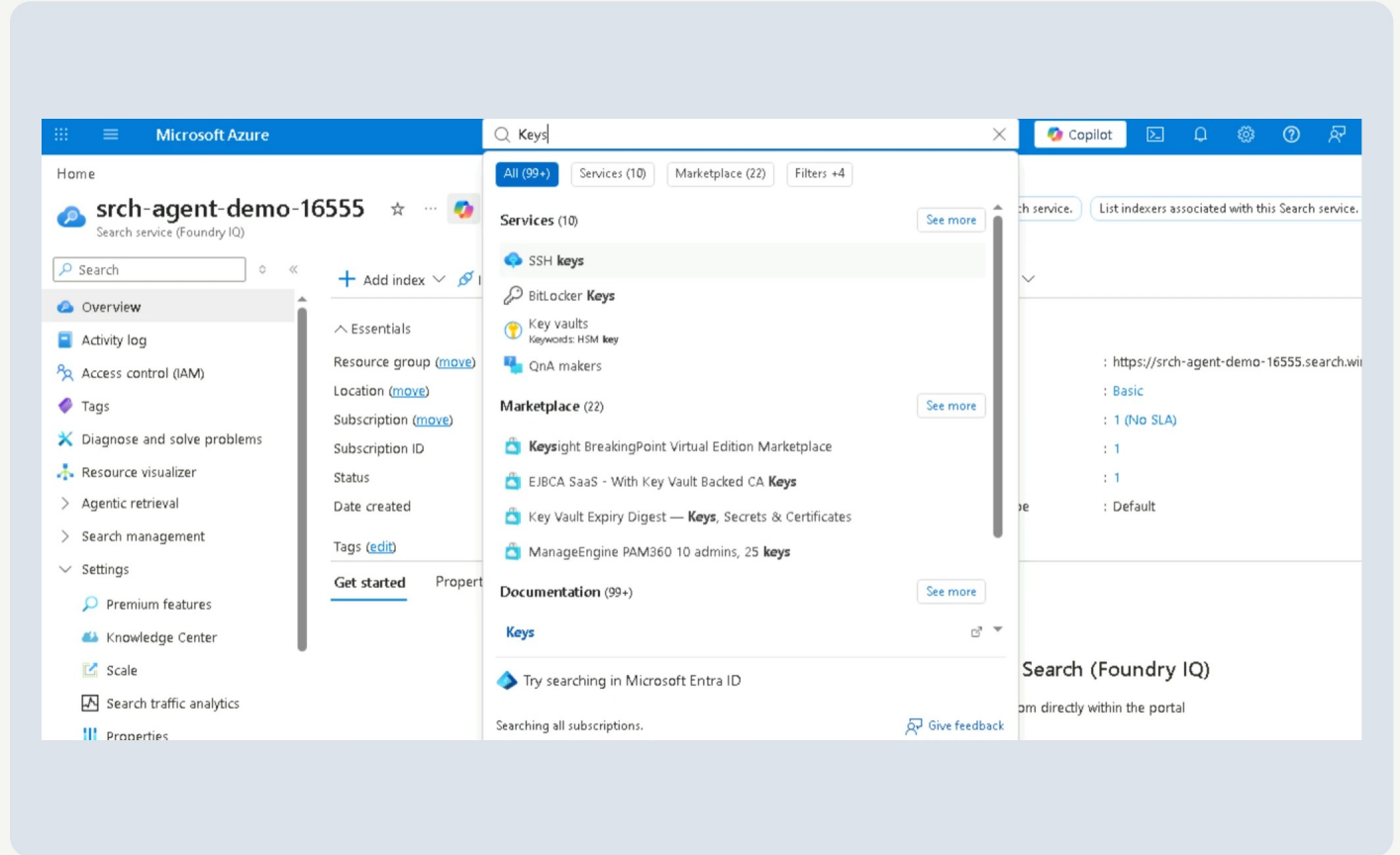
STEP 5 • AI SEARCH

Use the small search box, not the top bar

1. Typing “Keys” in the **big top bar** searches all of Azure: SSH keys, Key vaults... Don't click those.
2. Press **Esc**.
3. Use the **small box** on the left, above Overview. It filters this resource's menu.
4. Type **Keys** there, or expand **Security + networking**.

Tip

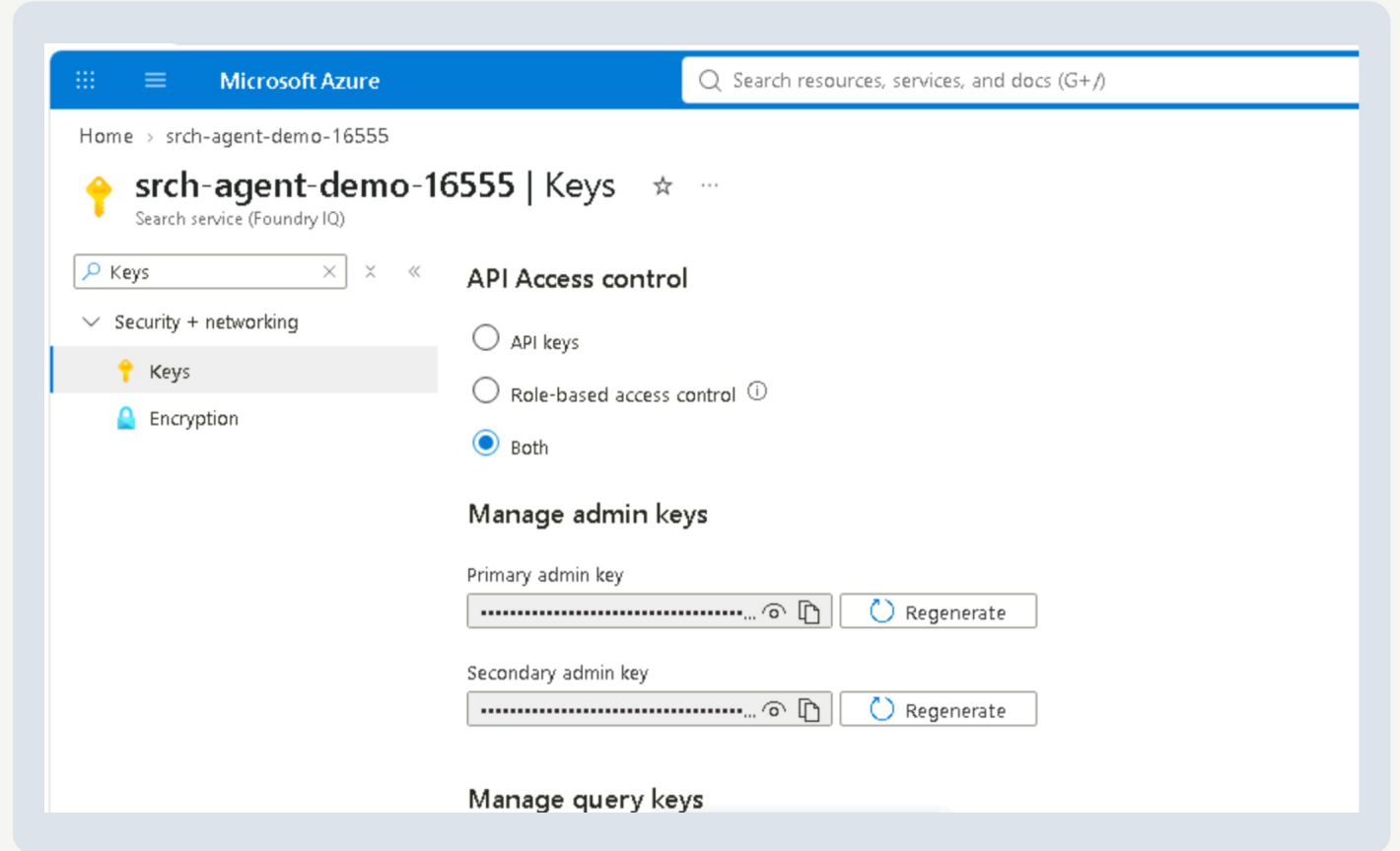
Big bar = find services. Small box = settings inside the resource you are in.



STEP 5 • AI SEARCH

Keys: allow login-based access

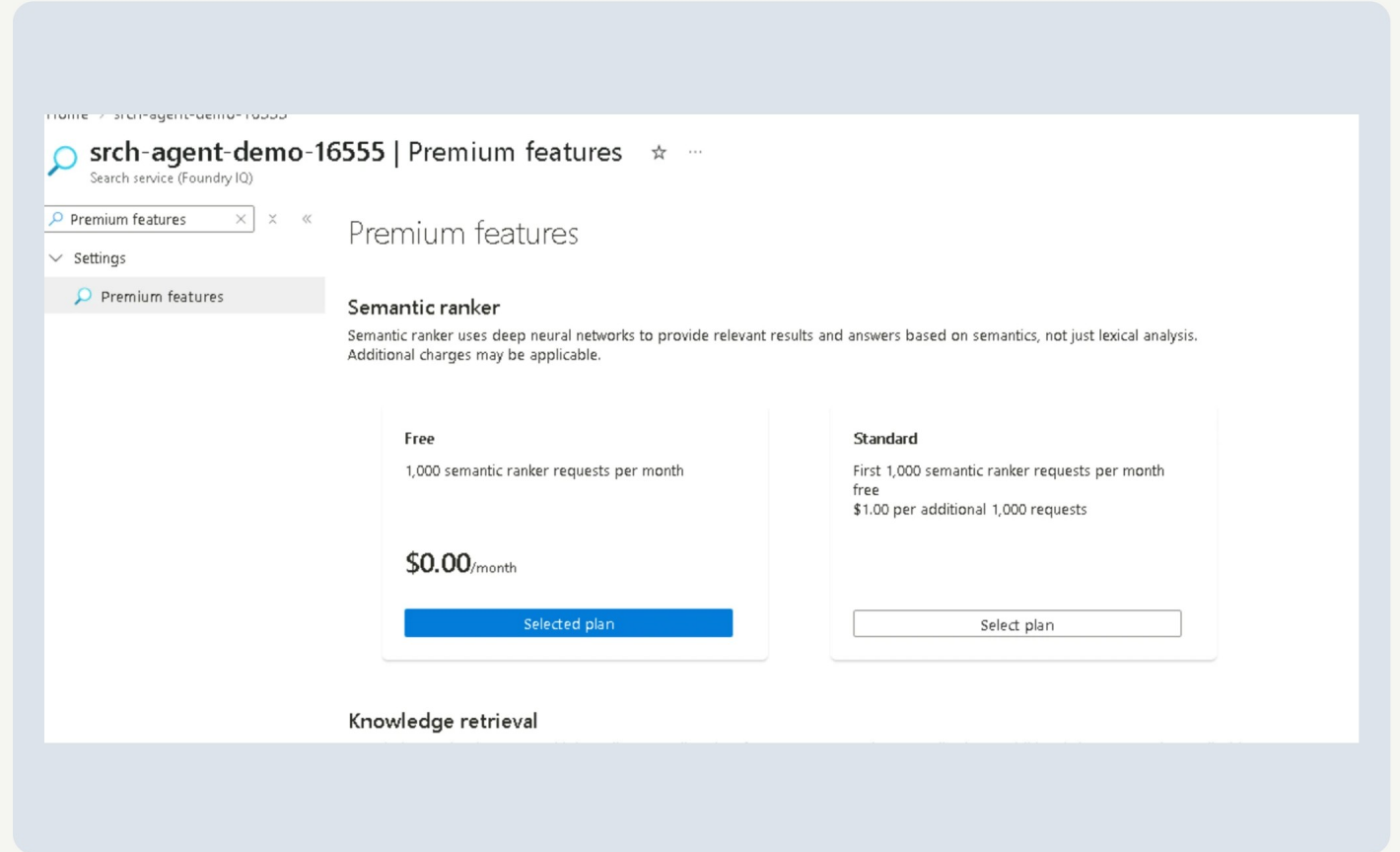
1. Small menu box: **Keys** (under Security + networking).
2. API Access control: **Both**.
3. It saves on click; confirm **Yes** if asked.
4. Do not copy the admin keys: the code signs in with your Azure login.



STEP 5 • AI SEARCH

Semantic ranker: Free plan

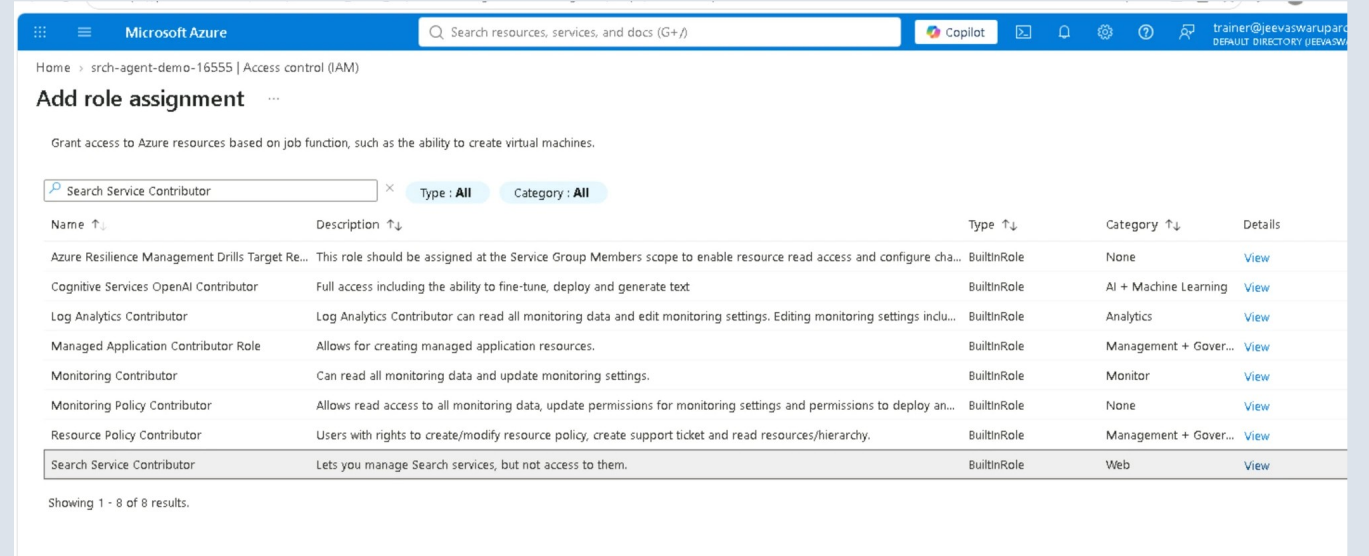
1. Small menu box: **Premium**, then **Premium features**.
2. Semantic ranker: **Free** shows “Selected plan”.
3. 1,000 requests a month at \$0 is plenty for the labs.
4. Leave Knowledge retrieval unchanged.



STEP 6 · SEARCH PERMISSIONS

Pick the role Search Service Contributor

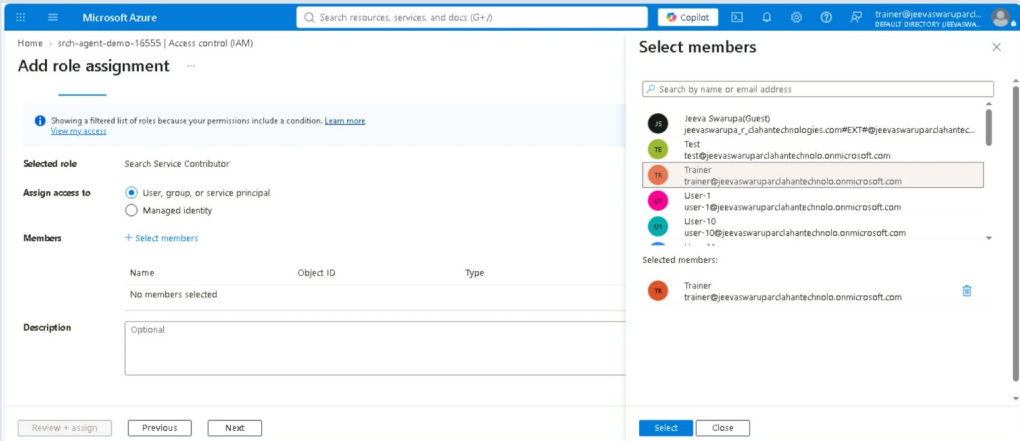
1. Small menu box: **Access control (IAM)**.
2. **Add role assignment**.
3. Search **Search Service Contributor**.
4. Click the **name** of the last row (not “View”), then **Next**.



STEP 6 · SEARCH PERMISSIONS

Select yourself, then Review + assign

+ Select members → Trainer → Select



Review + assign (scroll down if hidden)

Role	Members	Conditions	Review + assign
Role	Search Service Contributor		
Scope	/subscriptions/01ebc39d-a18d-4030-a585-9257e2b2d7e8/resourceGroups/rg-ai-agent-demo/providers/Microsoft.Search/searchServices/srch-agent-demo-16555		
Members	Name	Object ID	Type
	Trainer	8bfb7f0e-f8a0-427f-9836-6045a0a85529	User
Description	No description		

STEP 6 • SEARCH PERMISSIONS

Second role: Search Index Data Contributor

Same steps, this role

Add role assignment

RoleMembersConditionsReview + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)  Copilot can help pick a role

Job function rolesPrivileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Search Index Data Contributor

Type: AllCategory: All

Name	Description	Type	Category	Details
Search Index Data Contributor	Grants full access to Azure Cognitive Search index data.	BuiltInRole	Web	View

Showing 1 - 1 of 1 results.

Owner alone cannot read or write documents

Home>srch-agent-demo-16555

srch-agent-demo-16555 | Access control (IAM)

Search service (foundry IO)

Access

+ AddDownload role assignmentsEdit columnsRefreshDeleteFeedback

Access control (IAM)

Security + networkingNetworking

Check accessRole assignmentsRolesDeny assignments

Number of role assignments for this subscription58\$000

Search by name or emailType: AllRole: AllScope: All scopesGroup by: Role

All (38)Job function roles (1)Privileged administrator roles (37)

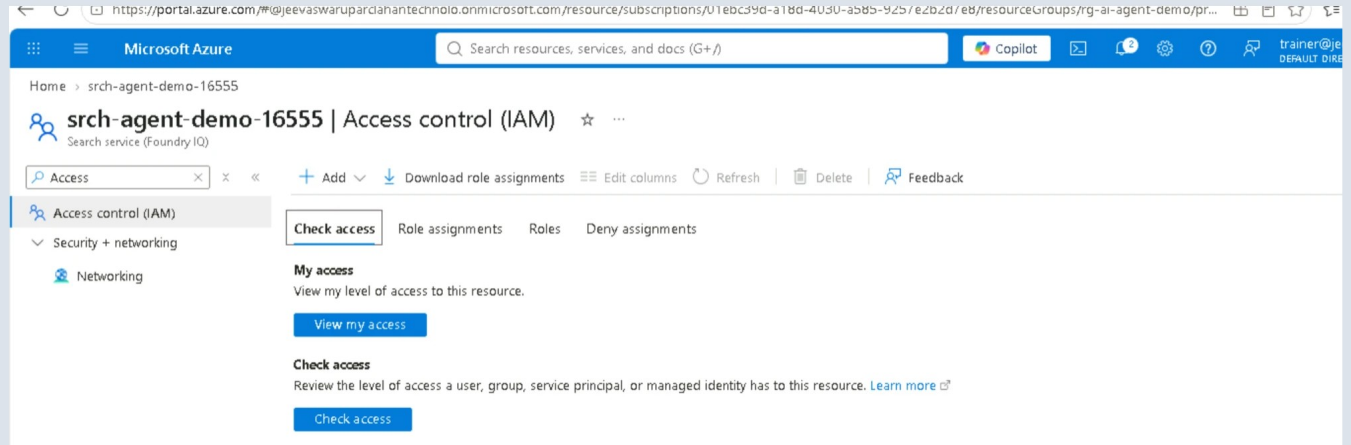
Name	Type	Role	Scope	Condition
Owner (19)				
☐ Heeva Swarnupa	User	Owner	Subscription (inherited)	None
☐ Test	User	Owner	Subscription (inherited)	View
☐ Trainer	User	Owner	Subscription (inherited)	View
☐ User-1	User	Owner	Subscription (inherited)	View

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STEP 6 · SEARCH PERMISSIONS

Search is finished

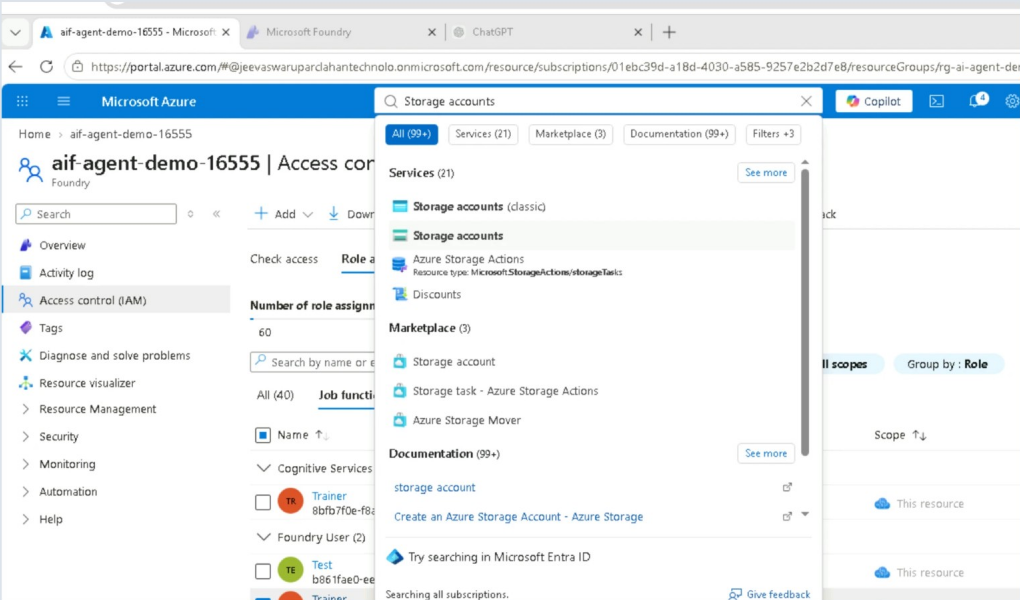
1. The bell shows one notification per role added.
2. **Role assignments** → **Job function roles** lists both roles for Trainer.
3. Keys → Both · Semantic ranker → Free · two roles: done.



STEP 7 · STORAGE ACCOUNT

Create the Storage account

Top bar: Storage accounts (not classic)



Name, East US 2, Standard, LRS

Instance details

Storage account name * ⓘ

Region * ⓘ
[Deploy to an Azure Extended Zone](#)

Primary service *

Performance * ⓘ
☒ Standard: Recommended for most scenarios (general-purpose v2 account)
☐ Premium: Recommended for scenarios that require low latency.

Redundancy * ⓘ

STEP 7 · STORAGE ACCOUNT

Primary service is required

Choose “Azure Blob Storage or Azure Data Lake Storage”

Primary service *

Choose preferred storage type

Azure Blob Storage or Azure Data Lake Storage

Azure Files

Other (tables and queues)

Performance * ⓘ

☐ Premium: Recommended for scenarios that require low latency.

Review: hierarchical namespace Disabled

Create a storage account

Basics

Advanced

Networking

Data protection

Security

Encryption

Tags

Review + create

View automation template

Basics

Subscription

Azure subscription 1

Resource group

rg-ai-agent-demo

Location

East US 2

Storage account name

stagentdemo16555

Primary service

Azure Blob Storage or Azure Data Lake Storage

Performance

Standard

Replication

Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace

Disabled

Enable SFTP

Disabled

Enable network file system v3

Disabled

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STEP 7 • STORAGE ACCOUNT

Check networking, then create

Public network access: Enabled

Enable Defender for Storage	Disabled
Networking	
Public network access	Enabled
Public network access scope	Enabled from all networks
Default routing tier	Microsoft network routing
Data protection	
Point-in-time restore	Disabled
Blob soft delete	Enabled
Blob retainment period in days	7
Container soft delete	Enabled
Container retainment period in days	7
Classic file share soft delete	Enabled
Classic file share retainment period in days	7
Versioning	Disabled
Blob change feed	Disabled

Deployment complete

stagentdemo16555_1790487466286 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

OverviewInputsOutputsTemplate

✔ Your deployment is complete

Deployment name : stagentdemo16555_1790487466286

Subscription : [Azure subscription 1](#)

Resource group : [rg-ai-agent-demo](#)

Start time : 9/27/2026, 5:44:49 AM

Correlation ID : 2bc642f5-f393-4d16-88e5-bbca84b2ef39

> Deployment details

< Next steps

Go to resource

STEP 8 • APPLICATION INSIGHTS

Create Application Insights (the eyes)

Name, East US 2, OTLP Off, default workspace

Application Insights

Monitor web app performance and usage

PROJECT DETAILS

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure subscription 1

Resource Group * ⓘ rg-ai-agent-demo
[Create new](#)

INSTANCE DETAILS

Name ⓘ appi-agent-demo-16555 ✓

Region * ⓘ (US) East US 2

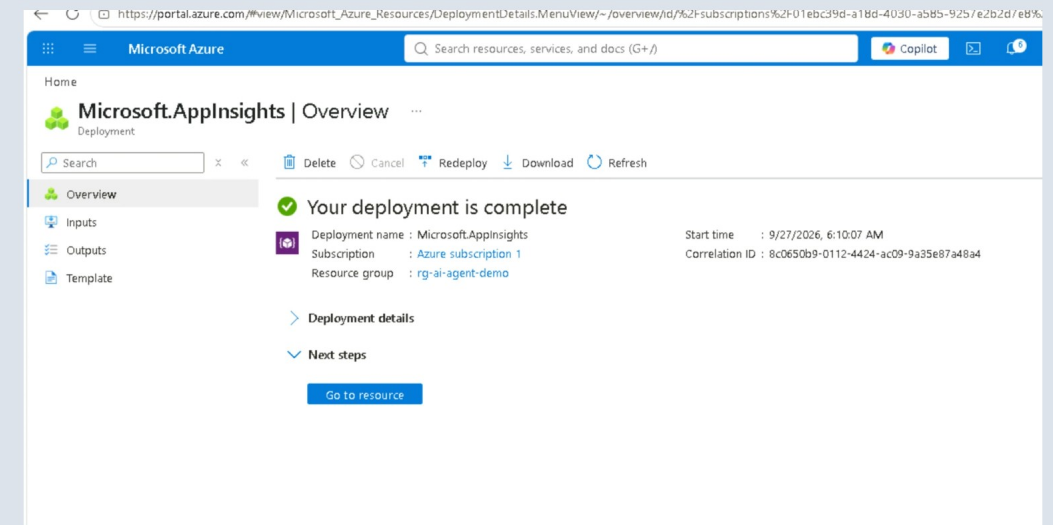
OTLP support ⓘ On **Off**

WORKSPACE DETAILS

Subscription * ⓘ Azure subscription 1

Log Analytics Workspace * ⓘ (new) DefaultWorkspace-01ebc39d-a18d-4030-a585-9257e2b2d7e8-EUS...

Deployment complete



STEP 9 · FUNCTION APP

Choose Flex Consumption

1. Top bar: **Function App**, then **+ Create**.
2. Select **Flex Consumption**: scales to zero, pay per use.
3. Not Consumption (Windows): no Python.
4. Not Premium or App Service: they bill even when idle.

Home > Function App

Create Function App

Select a hosting option

These options determine how your app scales, resources available per instance, and pricing. [Learn more about Functions hosting options](#)

Hosting plans	☒ Flex Consumption	☐ Functions Premium	☐ App Service	☐ Container Apps environment	☐ Consumption (Windows)
	Get high scalability with compute choices, virtual networking, and pay-as-you-go billing.	Deploy multiple function apps on the same plan with event-driven scaling.	Run web apps and function apps on the same plan with more compute choices and pay for the instances of the plan.	Host function apps with other containerized microservices and pay for compute capacity.	Pay for compute resources when your functions are running (pay-as-you-go).
Scale to zero	✓	-	-	✓	✓
Scale behavior	Fast event-driven	Event-driven	Metrics based	Event-driven with KEDA	Event-driven
Virtual networking	✓	✓	✓	✓	-
Dedicated compute and prevent	Optional with Always Ready	Minimum of 1 instance required	Minimum of 1 instance required	Optional with minimum replicas	-

Select

STEP 9 · FUNCTION APP

Basics: Python 3.11 in East US 2

1. Name **func-agent-demo-16555**, region **East US 2**.
2. Runtime **Python**, version **3.11**, instance **2048 MB**.
3. Zone redundancy: Disabled.
4. Note the unique web address under the name. It is not simply name.azurewebsites.net.

Home > Function App > Create Function App

Create Function App (Flex Consumption) ...

Manage all your resources.

Subscription * ⓘ Azure subscription 1

Resource Group * ⓘ rg-ai-agent-demo
[Create new](#)

Instance Details

Function App name * func-agent-demo-16555 ✓
-b7b5abgrc6g2g6ht.eastus2-01.azurewebsites.net

Region * East US 2

Runtime stack * Python

Version * 3.11

Instance size * ⓘ 2048 MB

Zone redundancy

Instances of your app are distributed across availability zones for increased reliability. [More about zone redundancy.](#)

[Review + create](#) < Previous Next : Storage >

STEP 9 · FUNCTION APP

Watch for “(New)”: pick the resources you already made

Before: two unwanted new resources

Hosting	
Storage (New)	
Storage account	rgaiagentdemoa88a
Plan (New)	
Hosting options and plans	Flex Consumption
Name	ASP-rgaiagentdemo-b7b0
Operating System	Linux
Region	East US 2
Monitoring (New)	
Application Insights	Enabled
Name	func-agent-demo-16555
Region	East US 2

Storage tab: choose stagentdemo16555

Create Function App (Flex Consumption) ...

Basics **Storage** Azure OpenAI Networking Monitoring Durable Functions Deployment Authentication Tags Review + create

Storage

Select a storage account or create a new one. Accounts must support blobs, queue, and Table storage. [Learn more](#)

Storage account * stagentdemo16555 (v2) [Create new](#)

Diagnostic Settings

Configure diagnostic settings to enable data monitoring for your storage account. [Learn more](#)

Blob service diagnostic settings

☒ Configure later (Recommended for custom controls) Select this option for control over log destinations, retention policies, and specific logs and metrics.

☐ Configure now (Recommended for basic controls) Select to configure Azure Log Analytics with storage, write logs, and transaction.

[Review + create](#) [< Previous](#) [Next : Azure OpenAI >](#)

Monitoring fixed, final review, create

Monitoring: appi-agent-demo-16555

Create Function App (Flex Consumption) ...

Basics Storage Azure OpenAI Networking **Monitoring** Durable Functions Deployment Authentication Tags Review + create

Application Insights
Azure Monitor application insights is an Application Performance Management (APM) service for developers and DevOps professionals. Enable it below to automatically monitor your application. It will detect performance anomalies, and includes powerful analytics tools to help you diagnose issues and to understand what users actually do with your app. Your bill is based on amount of data used by Application Insights and your data retention settings. [Learn more](#)

[App Insights pricing](#)

Enable Application Insights * ☐ No ☒ Yes

Application Insights *

appi-agent-demo-16555 (East US 2) [Create new](#)

Region

East US 2

Review: no “(New)” except the plan

Create Function App (Flex Consumption) ...

Runtime stack	Python 3.11
Instance size	2048 MB

Hosting

Storage	
Storage account	stagentdemo16555

Plan (New)

Hosting options and plans	Flex Consumption
Name	ASP-rgaigagentdemo-b7b0
Operating System	Linux
Region	East US 2

Monitoring

Application Insights	Enabled
Name	appi-agent-demo-16555
Region	East US 2

Create

< Previous

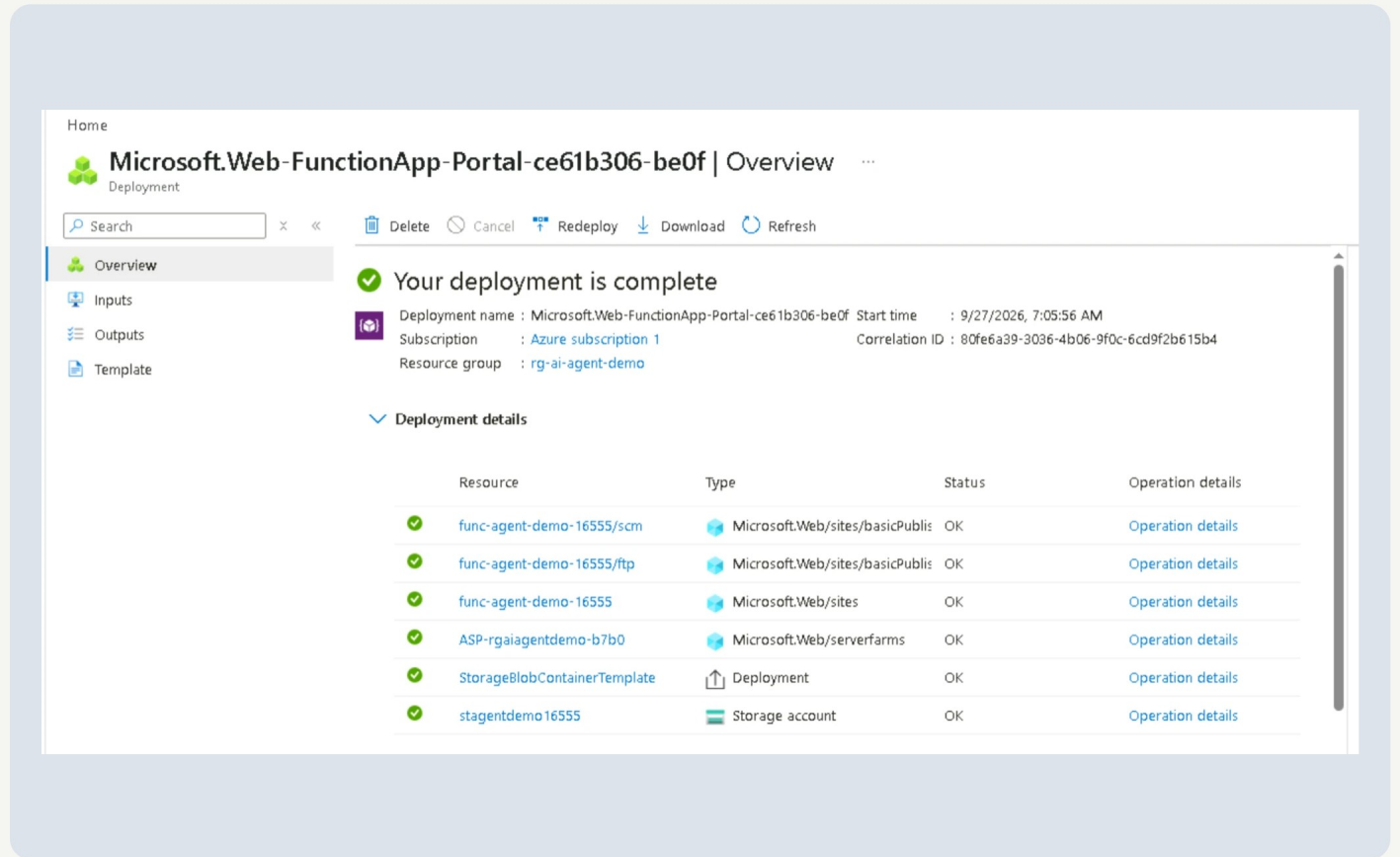
Next >

[Download a template for automation](#)

STEP 9 · FUNCTION APP

Created: now turn on its identity

1. All rows OK: the app, its plan, a code folder inside **your** storage account.
2. **Go to resource**, small menu box: **Identity**.
3. System assigned: **On**, **Save**, **Yes**.
4. An Object (principal) ID appears: the app now has its own login.



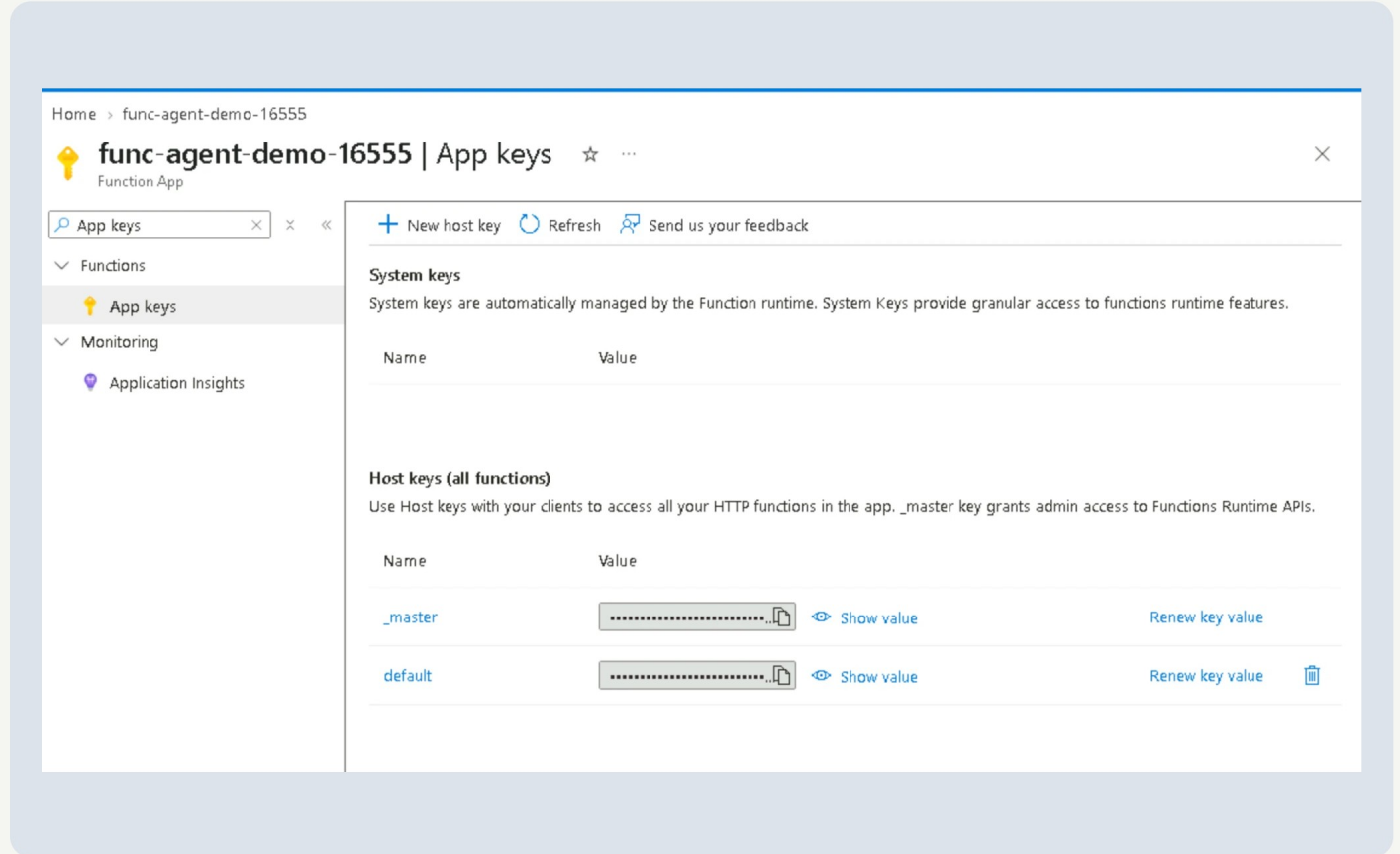
STEP 10 · KEYS

The default host key is FUNCTION_KEY

1. Small menu box: **App keys**.
2. Row **default**: click the copy icon.
3. Never use **_master**: it is admin access.
4. Needed only after the lab code is deployed to this app.

Watch out

The only real secret in .env. Not in chat, slides or Git. If it leaks: Renew key value.



The .env file on the client machine

Variable	Value / where it comes from
RESOURCE_GROUP	rg-ai-agent-demo
FOUNDRY_ENDPOINT	https://aif-agent-demo-16555.services.ai.azure.com/api/projects/proj-default
MODEL_DEPLOYMENT_NAME	gpt-5-mini
EMBEDDING_DEPLOYMENT_NAME	text-embedding-3-small
SEARCH_ENDPOINT	https://srch-agent-demo-16555.search.windows.net
SEARCH_SERVICE_NAME	srch-agent-demo-16555
FUNCTION_APP_NAME	func-agent-demo-16555
STORAGE_ACCOUNT_NAME	stagentdemo16555
APPLICATIONINSIGHTS_CONNECTION_STRING	copy from appi-agent-demo-16555 → Overview
FUNCTION_KEY	copy the default host key (secret)

STEP 11 · CLIENT MACHINE

Sign in, load the runbooks, test

```
az login --tenant jeevaswaruparclahantechnolo.onmicrosoft.com
az account set --subscription 01ebc39d-a18d-4030-a585-9257e2b2d7e8

cd day1 && python3 lab1_verify_environment.py
python3 lab4_build_search_index.py
cd ../day1-examples && python3 build_index_v2.py
python3 retrieval_examples.py
```

Tip

Lab 1 prints “environment ok” with latency and tokens: the model and your permissions work.

Watch out

A new Search service is empty. Until both index scripts run, every retrieval lab returns nothing.

REFERENCE

What went wrong, and the fix

Symptom	Cause	Fix
Region “out of the resources...”	No capacity for new Search services	Create again in East US (or Central US)
Role “Azure AI User” not found	Filtered role list for this account	Use Foundry User + Cognitive Services User
“Failed to add ... already exists”	Role was assigned automatically	Nothing to do
gpt-5-mini-2 appears	Deploy clicked twice	Delete the duplicate
429 Too Many Requests	Default 1K tokens per minute	Edit deployment → 100000
403 Forbidden from code	Role missing or not active yet	Check IAM; wait 5–10 minutes
Search returns nothing	Indexes not built	Run the two index scripts
“(New)” on Function App review	Storage/Monitoring tabs skipped	Pick your existing resources